

Waldemar

Claims + Evidence

PL Proofs

- logical
- computable / decidable
- $\Rightarrow$ constructive logic (intuitionistic)

Judgement (claim):

$\mathbb{Z} \text{ nat}$

Rules (evidence):

$\triangleright \text{data Nat} = \mathbb{Z}$
 $\triangleright \quad \quad \quad | \text{S Nat}$

$\triangleright \text{two} :: \text{Nat}$
 $\triangleright \text{two} = \text{S}(\text{S } \text{z})$

$\triangleright \text{three} = \text{S}(\text{two})$

$\frac{\text{z}}{\text{zero nat}}$

$\frac{\text{S} \quad \text{n nat}}{\text{succ n nat}}$

$\text{two} = \text{succ}(\text{succ zero})$

$\frac{\frac{\frac{\text{z}}{\text{zero nat}}}{\text{succ zero nat}}}{\text{succ (succ zero) nat}}$

Derivation

$\frac{\frac{\frac{\frac{\frac{\text{z}}{\text{zero nat}}}{\text{succ zero nat}}}{\text{succ (succ zero) nat}}}{\text{succ (succ (succ zero)) nat}}}{\text{succ (succ (succ zero)) nat}}$

Rules:

axiom

$\frac{\text{z}}{\text{zero nat}}$

$\frac{\text{S} \quad \text{n nat} \rightarrow \text{premise(s)}}{\text{succ n nat}} \rightarrow \text{conclusion(s)}$

$\triangleright \text{data Even} = \text{Zero}$
 $\triangleright \quad \quad \quad | \text{Even Odd}$

$\triangleright \text{data Odd} = \text{Odd Even}$

$\triangleright \text{one} :: \text{Odd}$
 $\triangleright \text{one} = \text{odd zero}$

$\triangleright \text{two} :: \text{Even}$
 $\triangleright \text{two} = \text{Even one}$

$\frac{\text{z}}{\text{zero even}} \text{ EVENZ}$

$\frac{\text{n odd}}{\text{succ n even}} \text{ EVEN}$

$\frac{\text{n even}}{\text{succ n odd}} \text{ ODD}$

$\frac{\frac{\frac{\text{z}}{\text{zero even}}}{\text{succ zero odd}}}{\text{succ (succ zero) even}} \text{ EVEN}$

Induction

$$\frac{z}{\text{zero nat}} \quad \frac{n \text{ nat}}{\text{succ } n \text{ nat}} s$$

P is property that holds for all nats if:

- It holds for zero ($P(\text{zero})$), and
- whenever it holds for n ($P(n)$), it also holds for $\text{succ } n$ ($P(\text{succ } n)$)

$$\frac{}{\text{zero even}} \text{ GIVEN } z \quad \frac{n \text{ odd}}{\text{succ } n \text{ even}} \text{ EVEN}$$

$$\frac{n \text{ even}}{\text{succ } n \text{ odd}} \text{ ODD}$$

Induction principle:

Let P be a property of even numbers
Let Q be a property of odd numbers

P/Q will hold for all even/odd numbers if:

- $P(\text{zero})$
- whenever n even and $P(n)$ then we have $Q(\text{succ } n)$
- whenever n odd and $Q(n)$, we have $P(\text{succ } n)$

Recipe for proof by induction:

① Decide who to induct on:

- You must induct on a premise
- Pick the "most interesting" premise

If x and y then z

(• Don't worry too much, you can always change your mind)

② State that you are doing a proof by induction and on which premise (readability)

③ Complete your proof by cases

- One case per rule / point of induction principle
- Assume premises
- Write down your IH